

Foundations of Biomedical Science

Image Analysis Workshop

10/30-11/06 2026

Iana Vasylieva, PhD

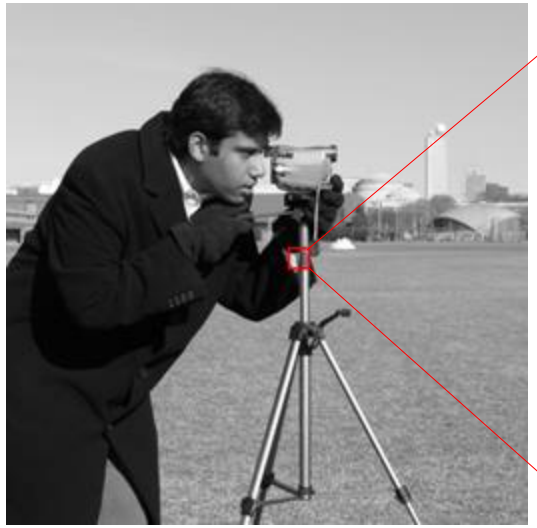
Learning objectives

1. Understand microscopy images as arrays of numerical pixel values.
2. Work with images that contain multiple channels, optical sections, and time points.
3. Gain an overview of image analysis software.
4. Familiarize with Python / Jupyter notebooks
5. Connect image processing / analysis operations into a workflow.
6. Have fun!

What Is a Digital Image?

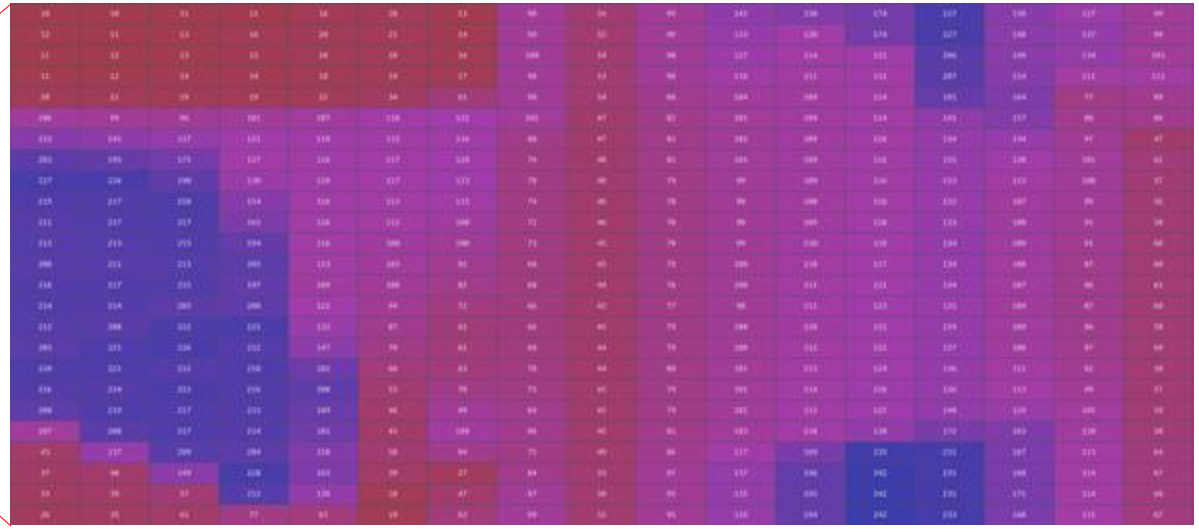
A gray-scale (one-channel) digital image is a 2D array of numbers (matrix, table).

Intensity pattern



People see at this level

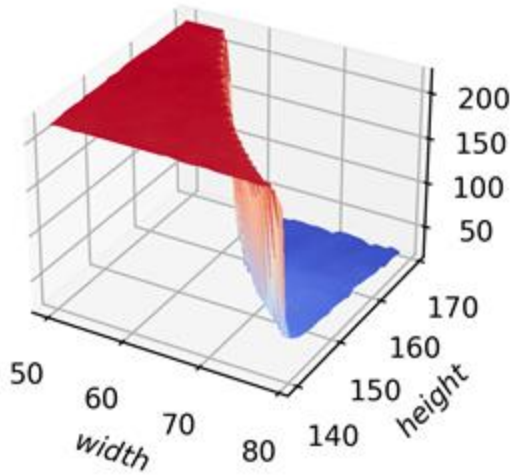
Matrix of integer numbers



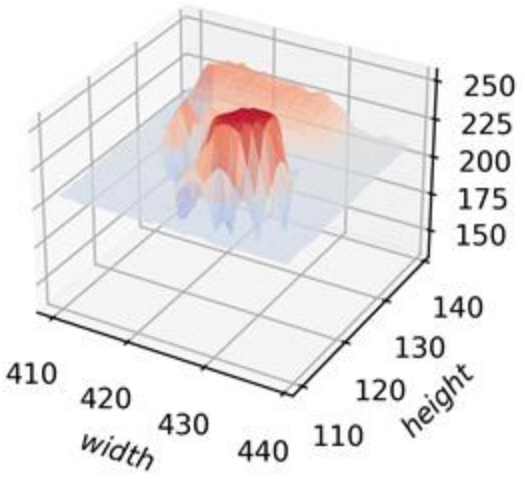
Computers “see” at this level

Gray-scale images









Color images

Color image – image defined in a color space.

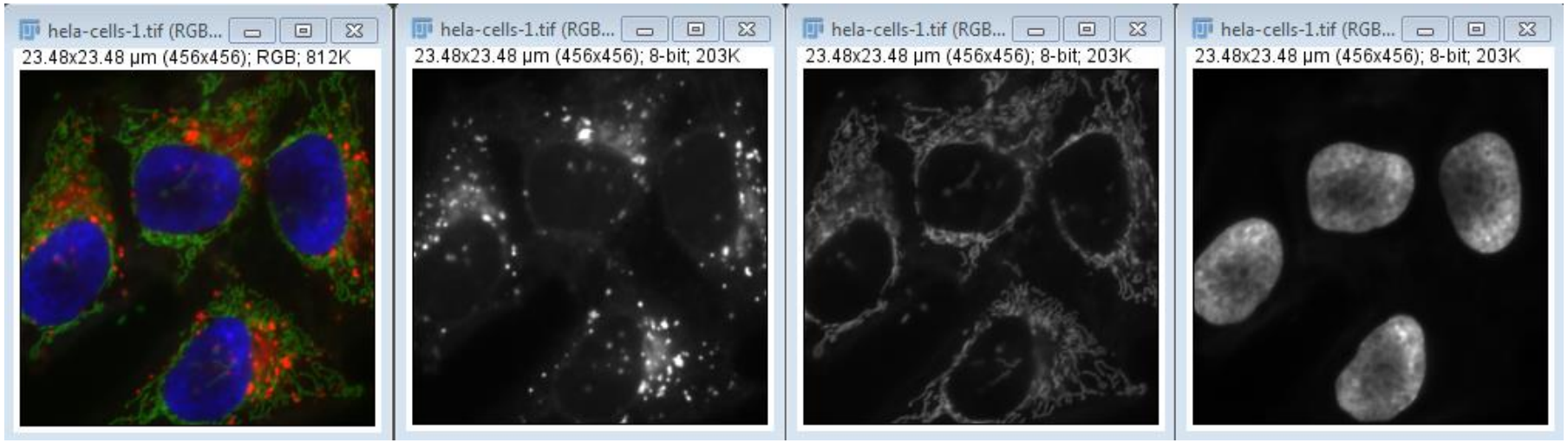
Color space is usually 3 dimensional like:

RGB (R – red, G – green and B – blue) or

CMY (C – cyan, M – magenta, Y – yellow) or

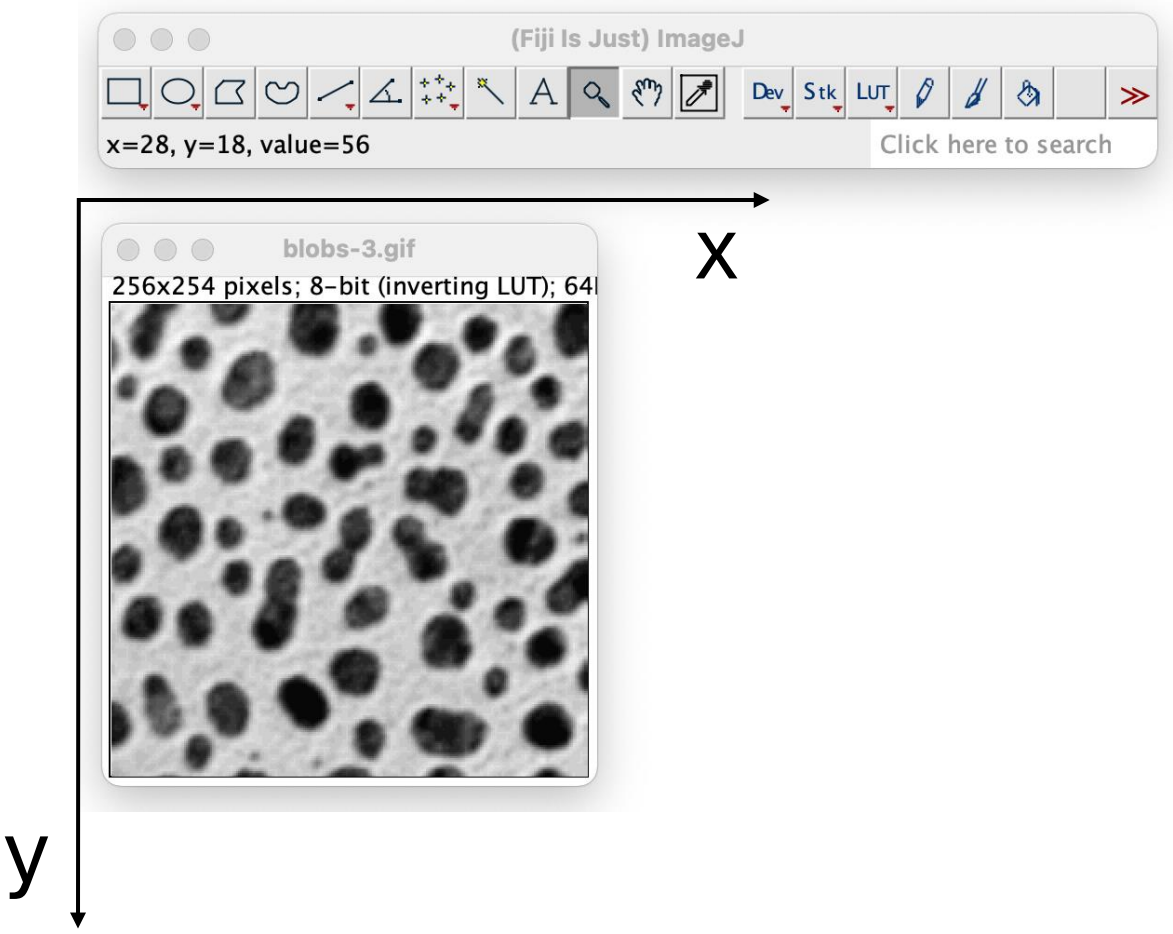
HSI (H – hue, S – saturation, I – intensity)

Color picker link



color red green blue

Coordinate system of images



Array

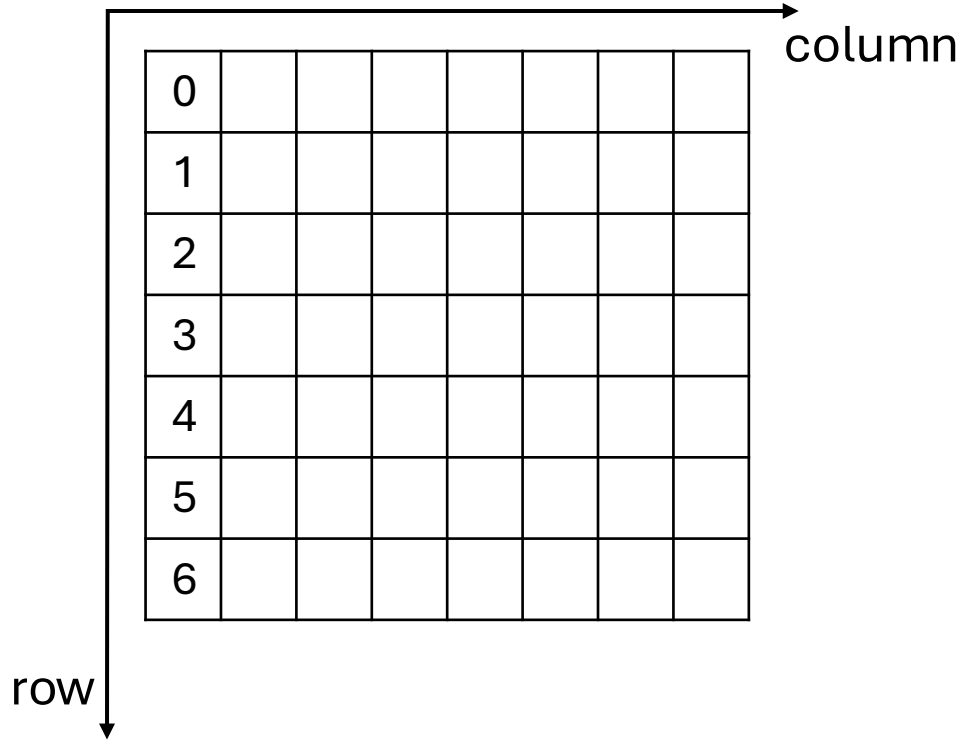


Image Spatial Resolution

Dots per inch (dpi) is a usual measure of image spatial resolution.

dpi = 25



75x75 pixels
7.2 kB (7,221 bytes)

dpi = 75



150x150 pixels
23.3 kB (23,347 bytes)

dpi = 300



900x900 pixels
524.2 kB (524,223 bytes)

All images are 8-bit.

Bit depth (intensity resolution)

2 bits (4 levels)



4 bits (16 levels)



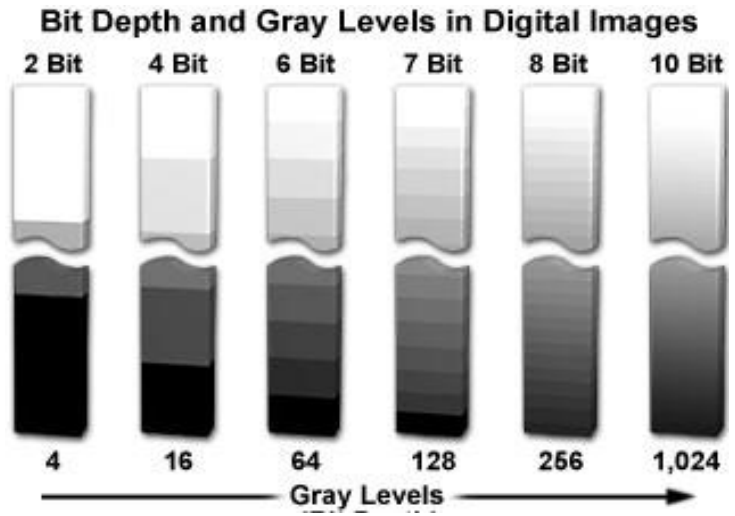
8 bits (256 levels)



All images are 900x900 px

Bit depth – number of bits used to represent the value of a single pixel

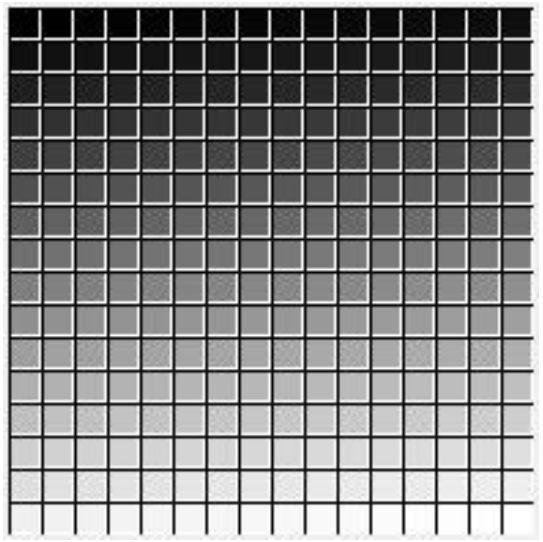
- 1 bit (2^1 = 2 gray levels) –> 0 – 1 (binary image)
- 8 bit (2^8 = 256 gray levels) –> 0 – 255
- 12 bit (2^{12} = 4096 gray levels) –> 0 – 4095
- 16 bit (2^{16} = 65,536 gray levels) –> 0 – 65,535
- 32 bit, 64 bit -> floating point (usually between 0 and 1)



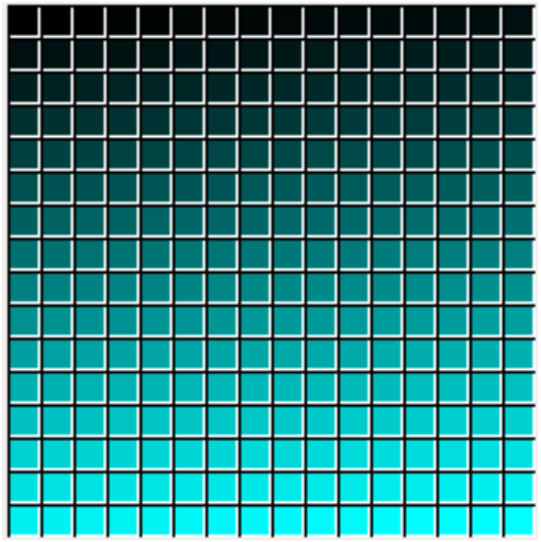
Source: https://micro.magnet.fsu.edu/primer/digitalimaging/digitalimagebasics.html

Gray images don’t need to be gray

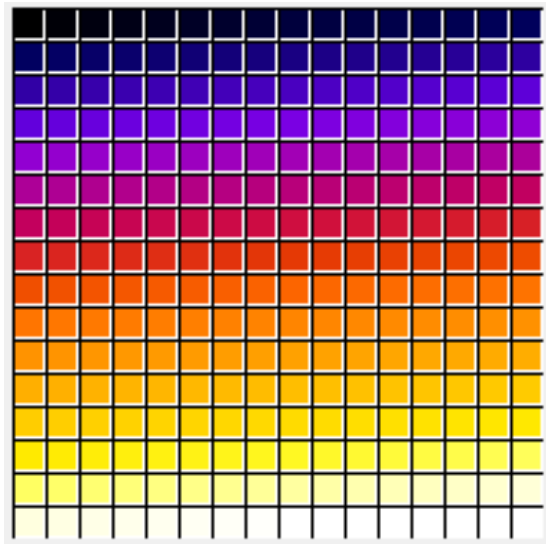
Look-up table (LUT) – (color table, palette) a function to assign different pseudo colors to a single channel image.



Gray LUT



Cyan LUT



Fire LUT

Gray images don’t need to be gray

Look-up table (LUT) – (color table, palette) a function to assign different pseudo colors to a single channel image.



Gray LUT



Cyan LUT

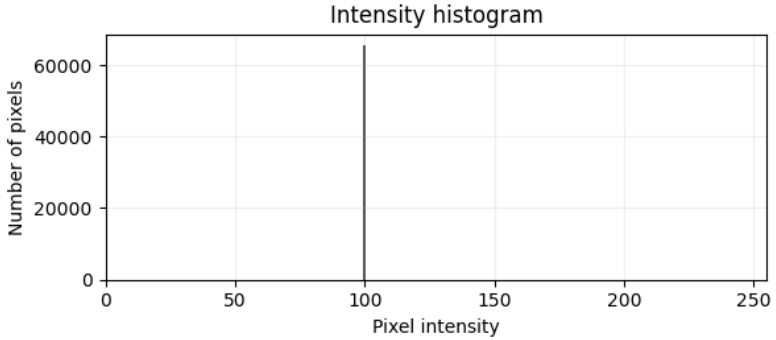


Fire LUT

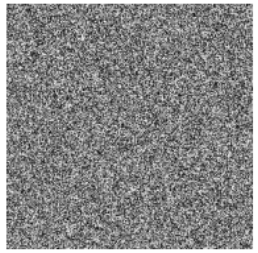
Image histogram

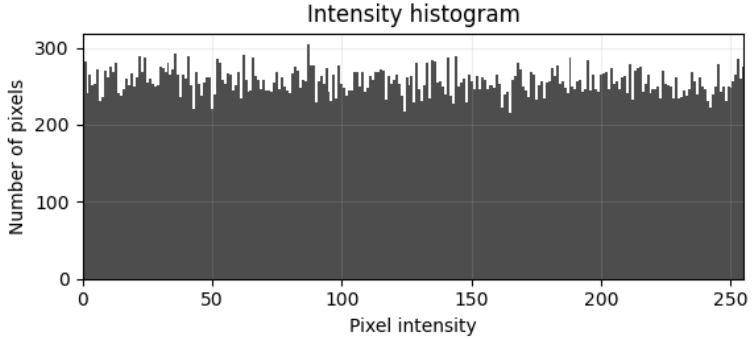
Constant intensity



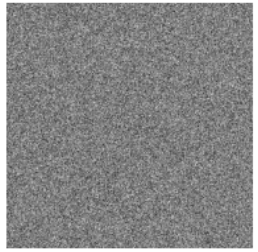


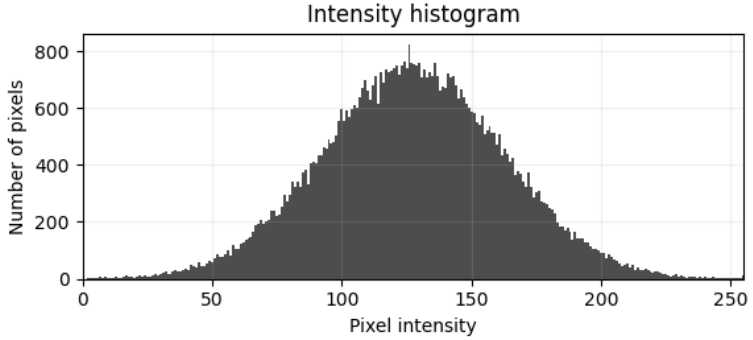
Uniform white noise





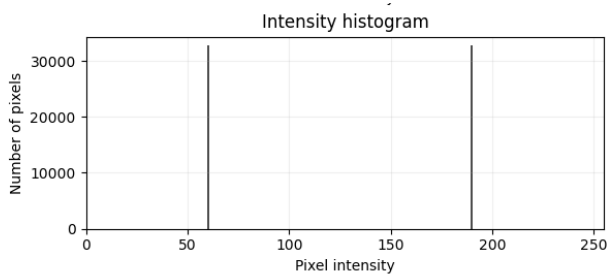
Gaussian white noise



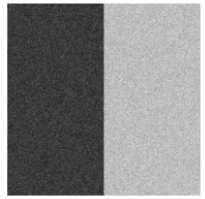


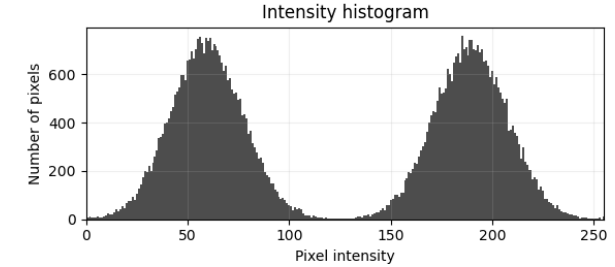
Two intensity levels





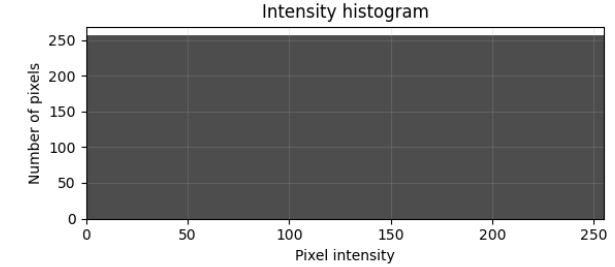
Two levels + noise



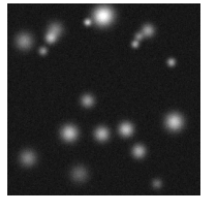


Intensity gradient





Fluorescent blobs



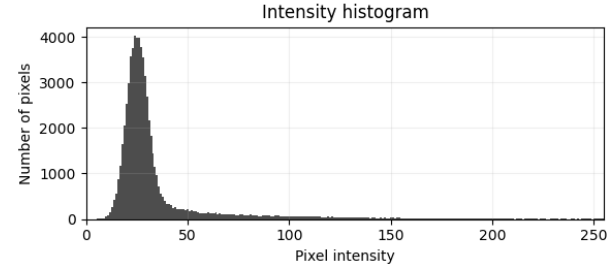


Image Visualization vs Image Processing vs Image Analysis

Image Visualization

Goal: help humans see and explore the image

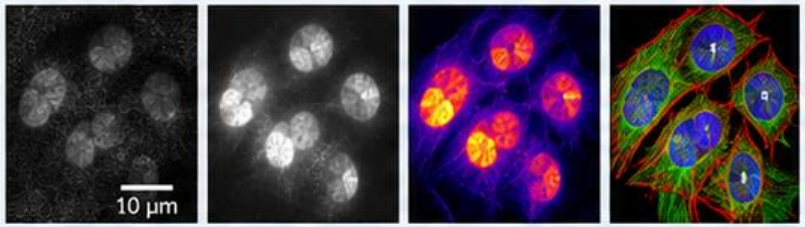
What it changes: appearance only, not the underlying data

Typical operations: brightness/contrast, LUTs, pseudocolor, overlays, zoom, 3D rendering

Output: a clearer view for inspection and presentation

Example questions: What structures are visible? Where is the signal?

Example: same image with different visualization settings



Original (grayscale) | Higher contrast | Pseudocolor (LUT) | Color overlay (multi-channel)

Image Processing

Goal: improve, transform, or prepare the image

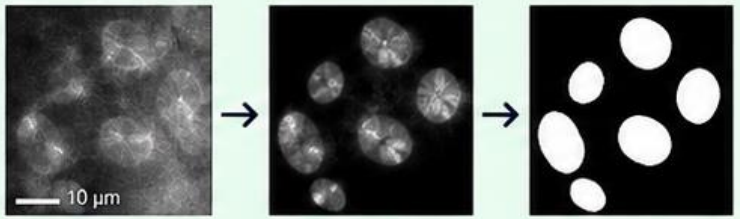
What it changes: the image data itself

Typical operations: denoising, background subtraction, filtering, deconvolution, registration, thresholding

Output: a modified image or mask ready for further use

Example questions: How can I clean up the image? How can I enhance signal?

Example: processing workflow



Raw image (noisy, uneven background) → Filtered / background corrected → Binary mask (thresholded)

Image Analysis

Goal: extract measurements, objects, patterns, or decisions

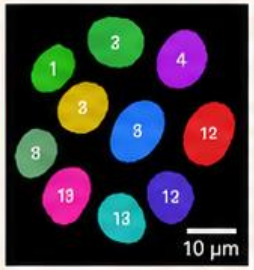
What it changes: produces quantitative results from the image

Typical operations: segmentation, connected components, feature extraction, tracking, classification, colocalization

Output: numbers, tables, labels, statistics, graphs, biological conclusions

Example questions: How many cells are there? What is their size, shape, or intensity?

Example: object analysis and quantitative results

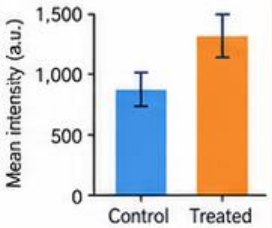


Segmented and labeled objects

Example measurement table

ID	Area (µm²)	Mean Intensity
1	52.3	1,240
2	48.1	980
3	66.5	1,305
4	59.2	1,102
5	47.6	876
...	...	...

Example quantitative result



Visualization = seeing
Helps you explore and understand the image (and supports all steps).

→ **Processing = transforming**
Improves, transforms, or prepares the image (often before analysis).

→ **Analysis = measuring/interpreting**
Extracts quantitative information and leads to biological insights.

Generated by ChatGPT 5

Image processing (image -> image)

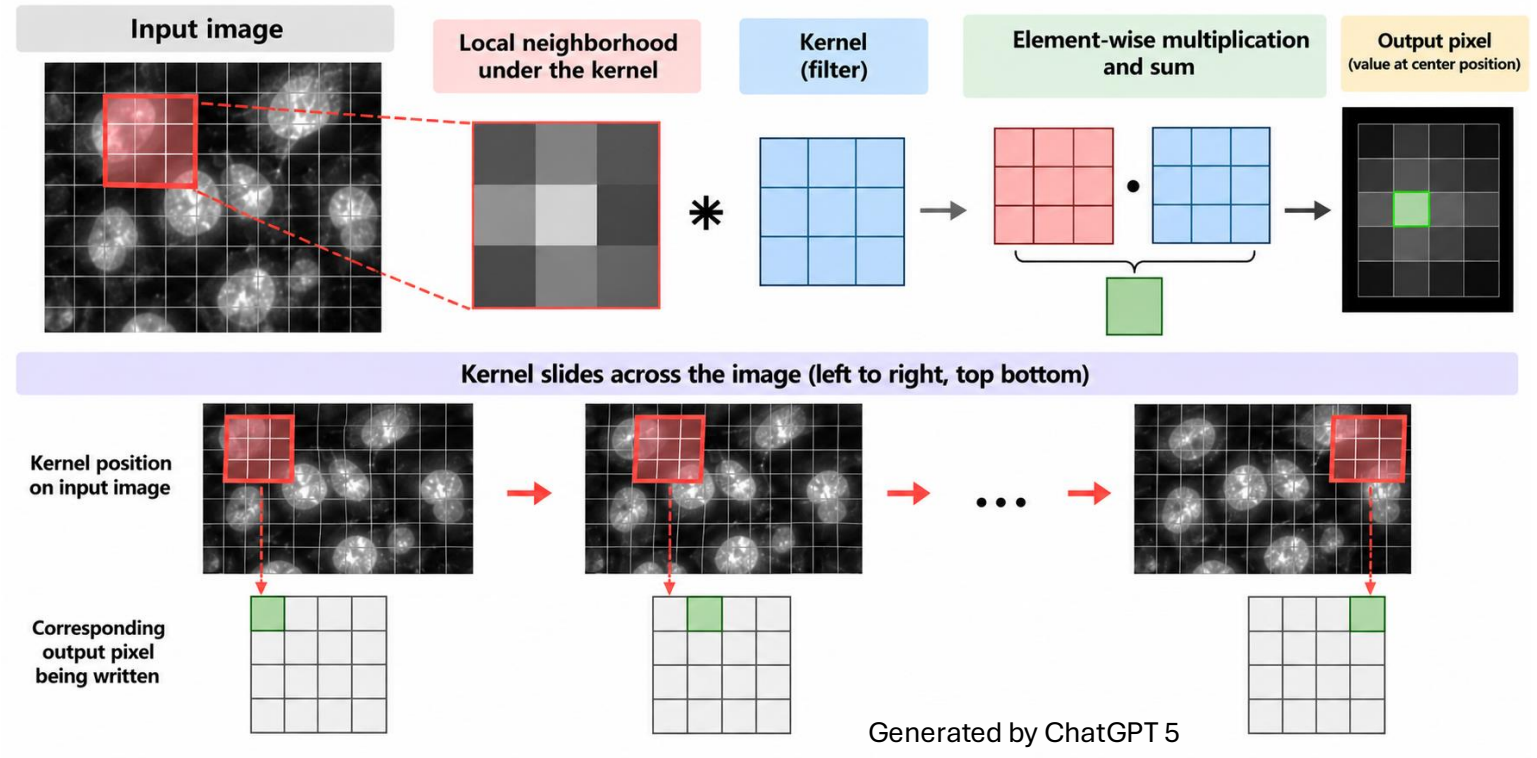


Image filtering / kernel operations

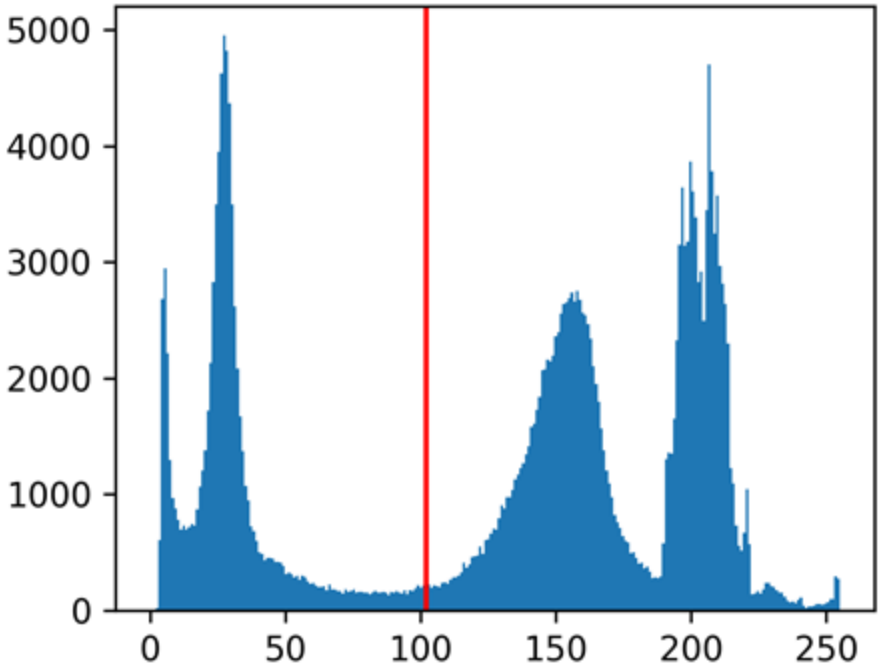
Smoothing / Blurring:	Mean (box) filter, Gaussian filter
Edge Detection:	Sobel, Prewitt, Scharr, Laplacian
Sharpening:	High-pass filter, Unsharp masking
Noise Removal:	Median filter, Bilateral filter, Non-local means
Morphological Operations:	Erosion, Dilation, Opening, Closing, Top-hat
Feature Detection:	Laplacian of Gaussian (LoG), Difference of Gaussians (DoG), Gabor filters

Thresholding / binarization

Original



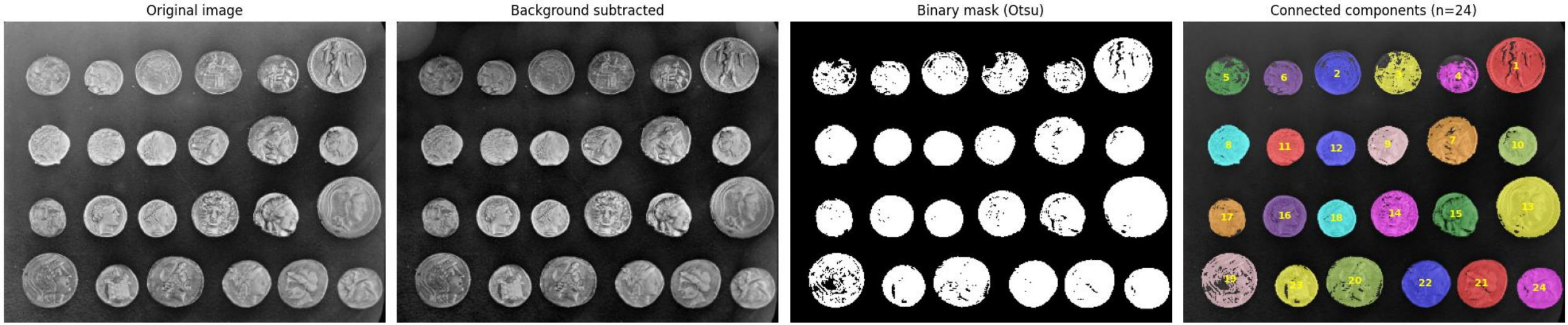
Histogram



Otsu Binary Mask

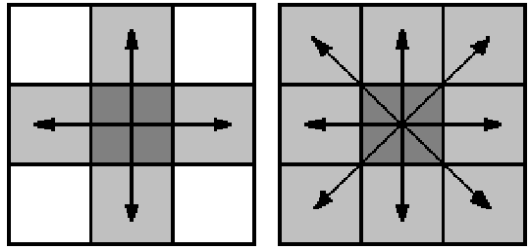


Image analysis (image -> data)



Thresholding: Converts grayscale intensities into a binary image (foreground = 1, background = 0).
Connected components: Groups neighboring foreground pixels into separate objects.
Labeling: Assigns a unique integer to each connected component (0 is reserved for background).

4-connectivity



8-connectivity

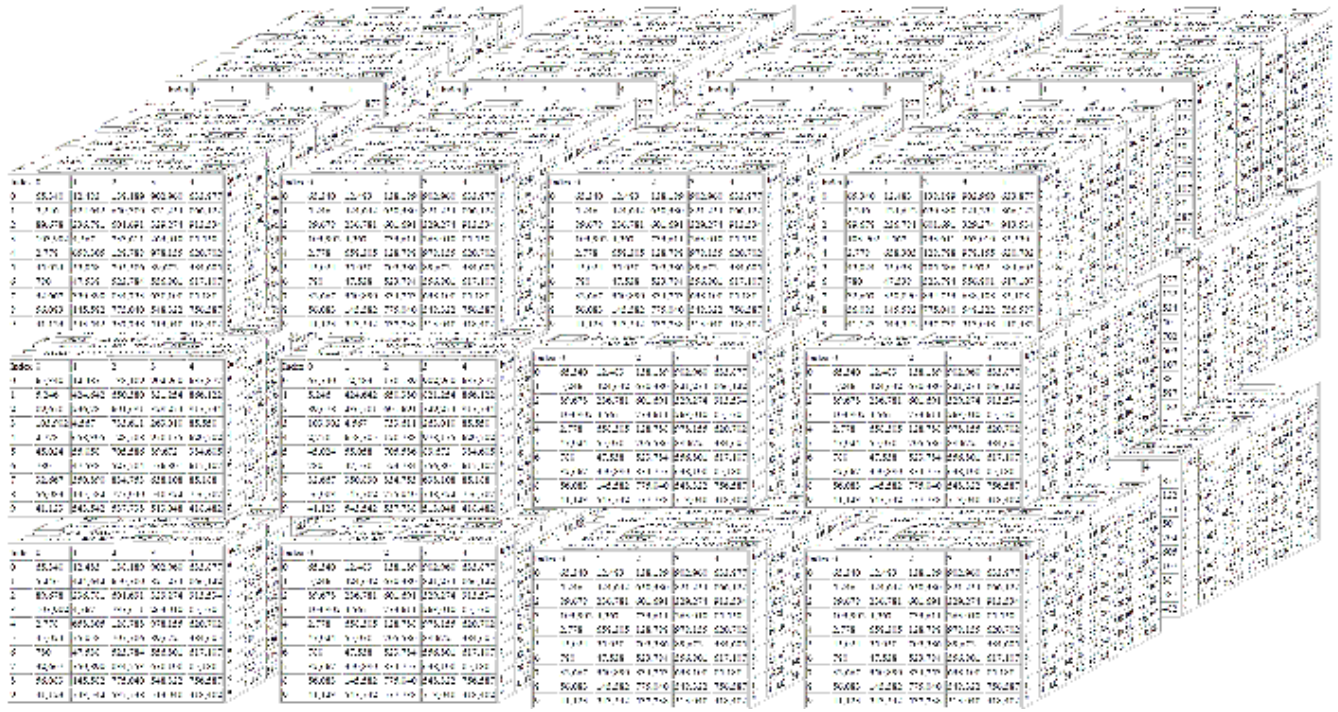
Bioimages are often multidimensional

What are different dimensions common for microscopy images?

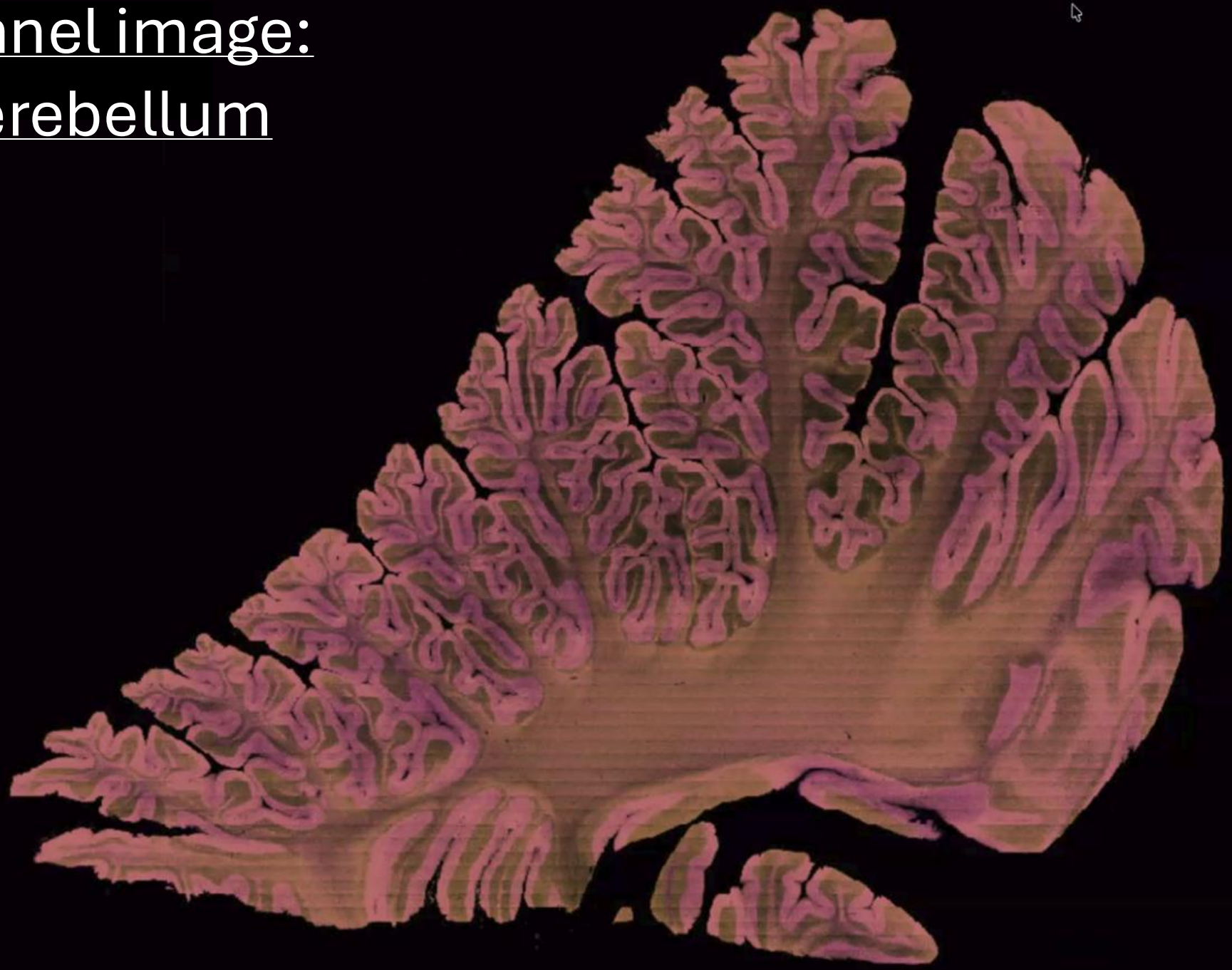
- different channels/colors, for example:
  - different fluorescence channels imaged simultaneously
  - different markers imaged sequentially
  - different modes of imaging

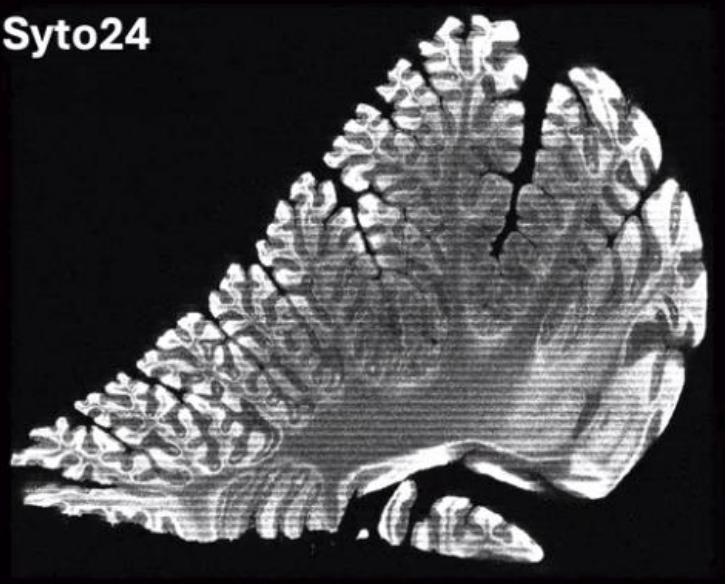
- time

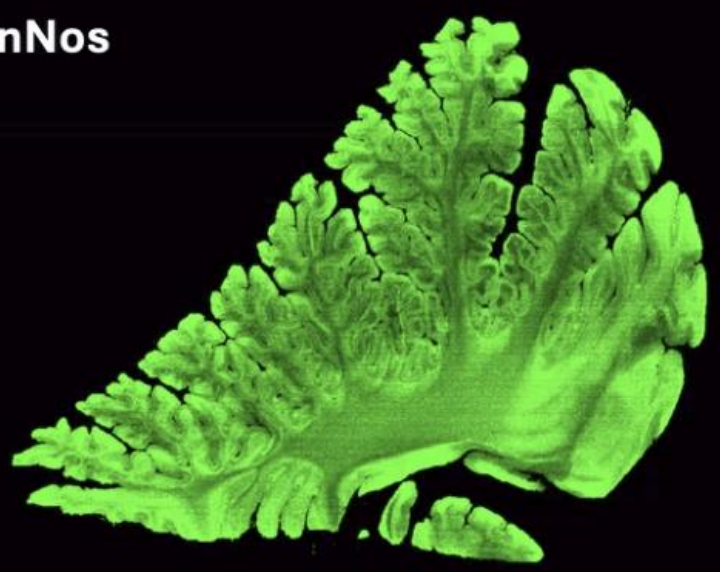
- depth in a sample

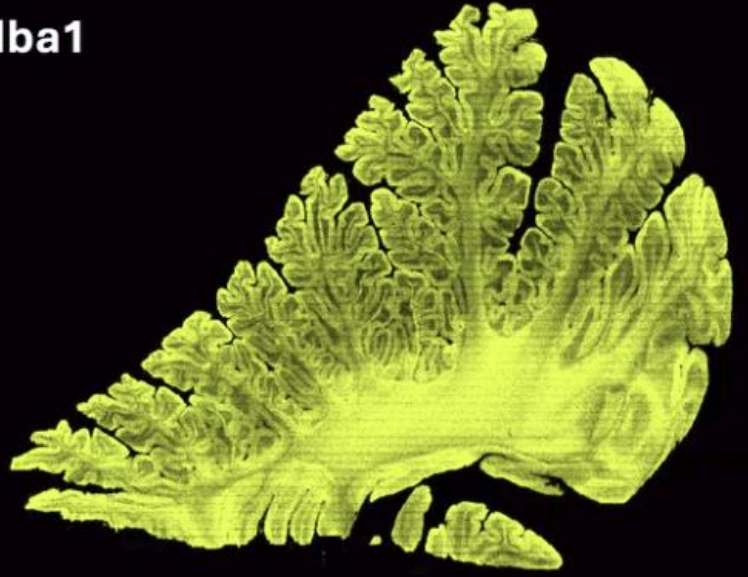


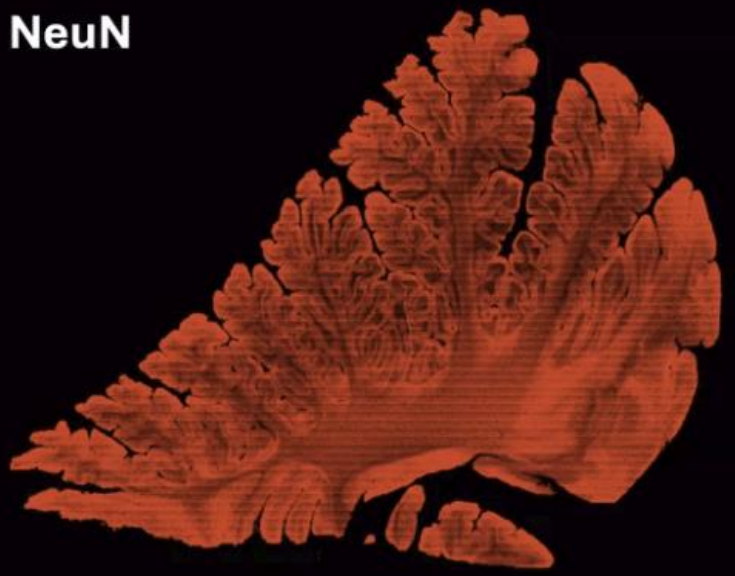
Multichannel image: human cerebellum

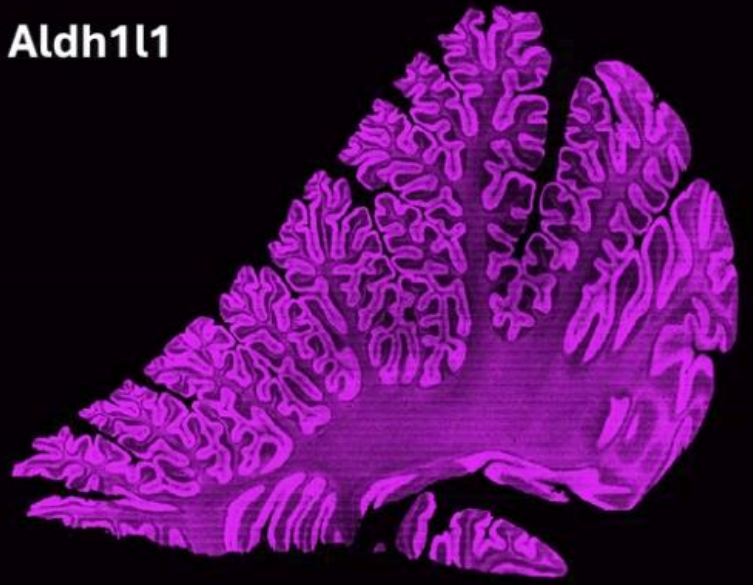


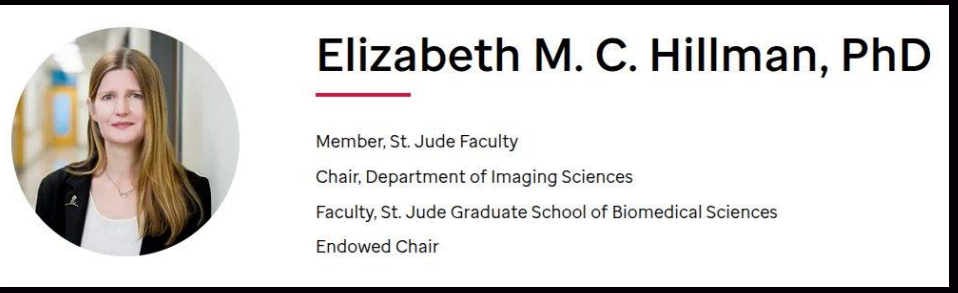












Elizabeth M. C. Hillman, PhD

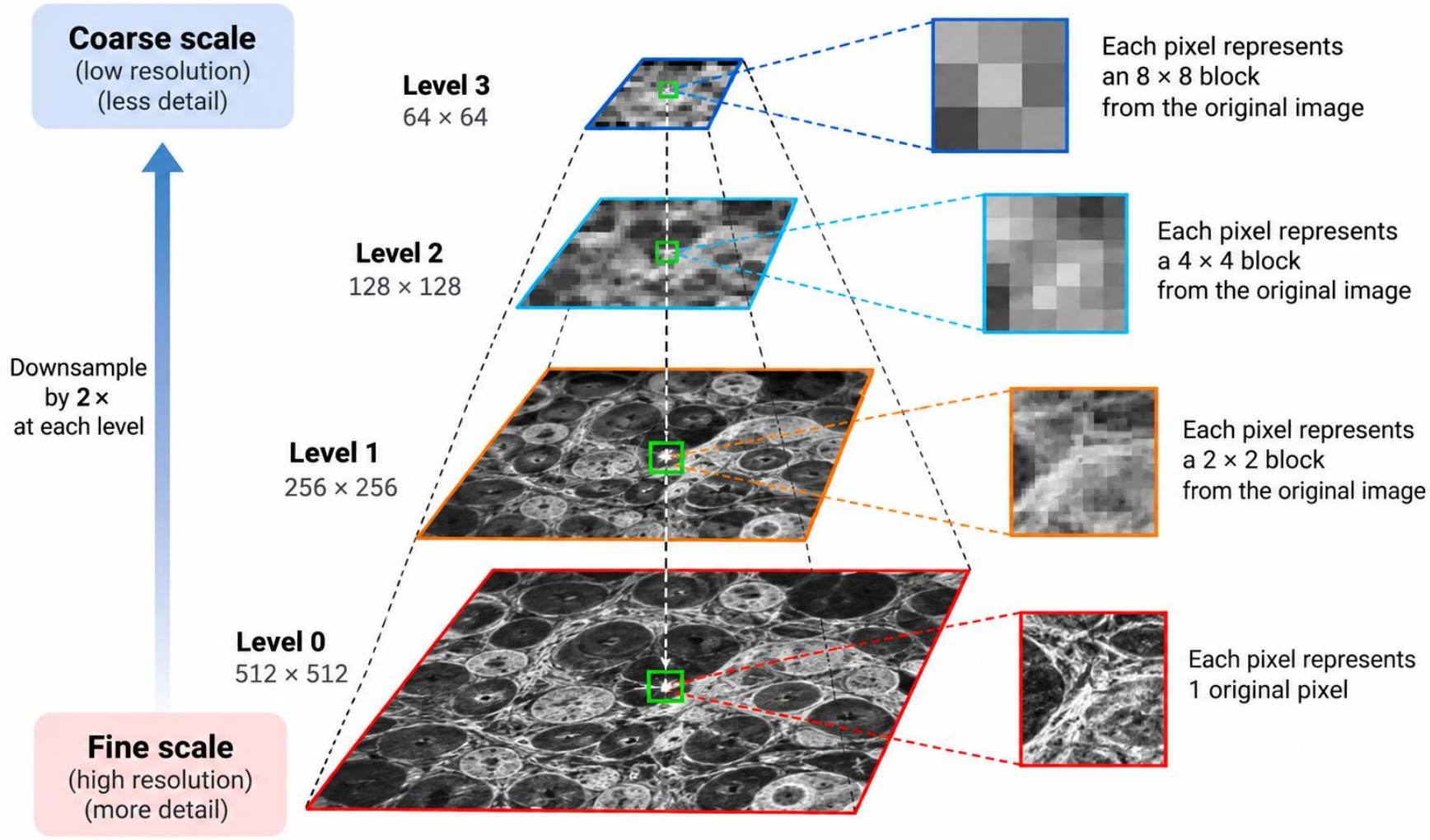
Member, St. Jude Faculty

Chair, Department of Imaging Sciences

Faculty, St. Jude Graduate School of Biomedical Sciences

Endowed Chair

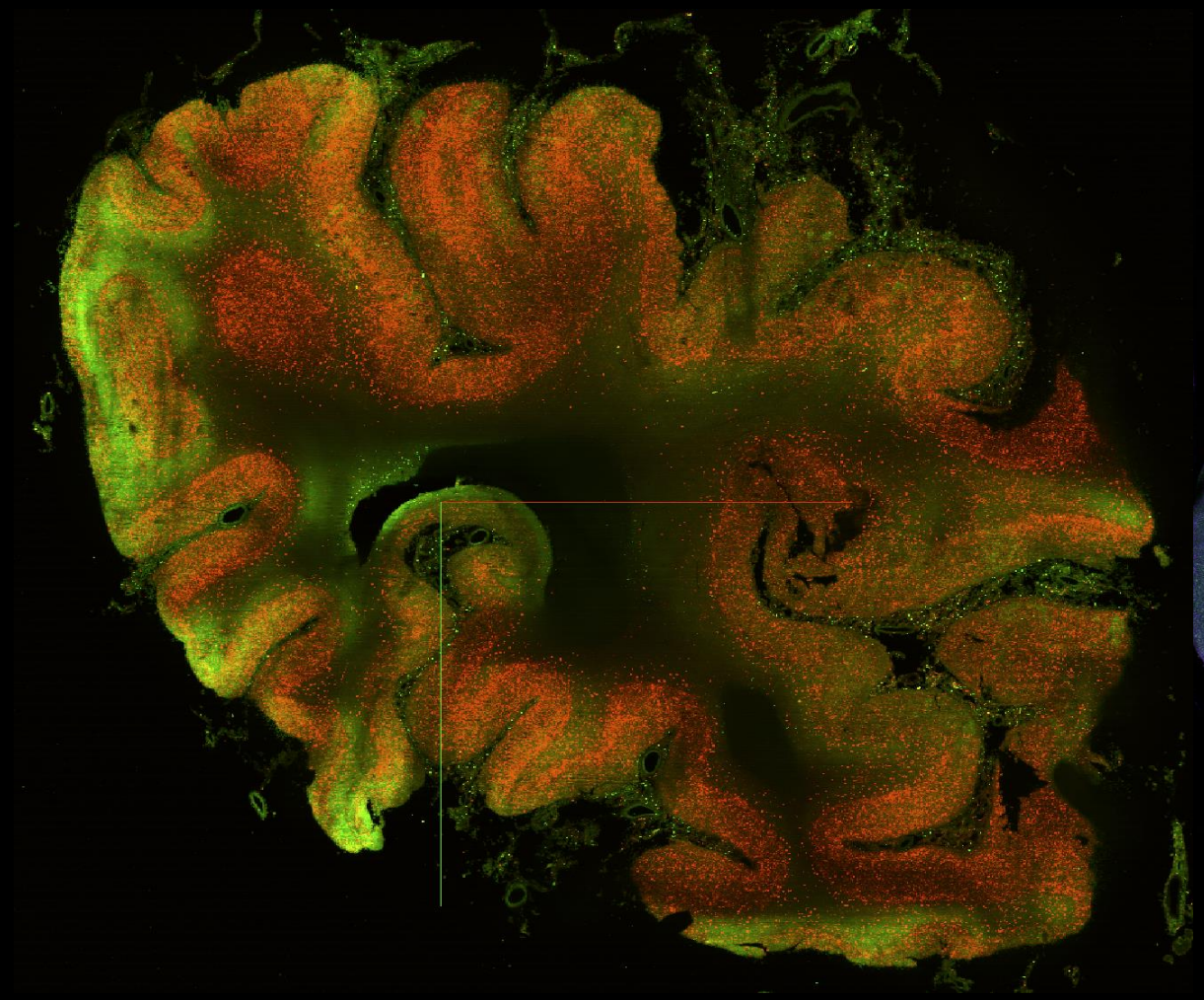
Multiscale image / pyramid



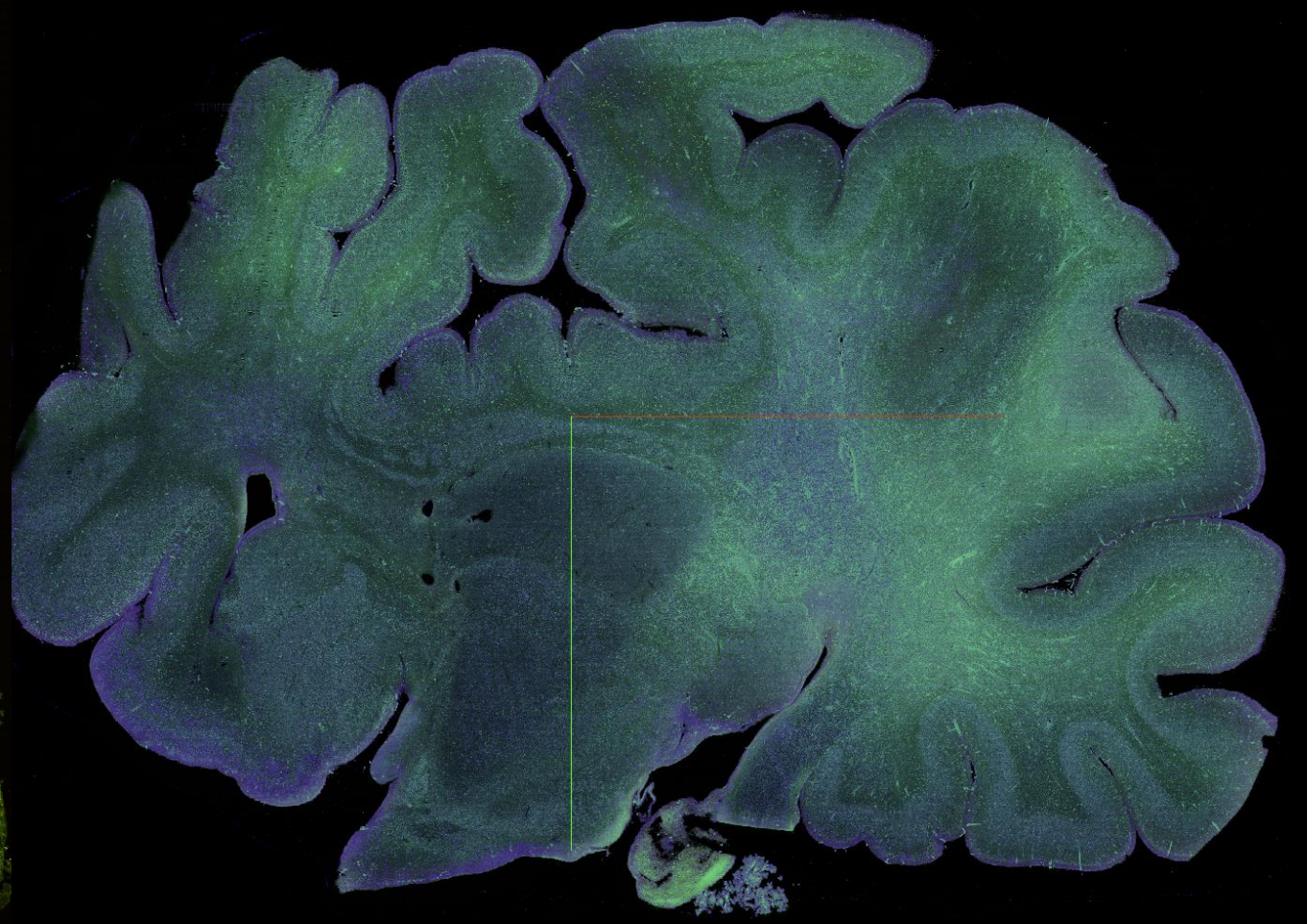
Generated by ChatGPT 5

Examples of multi-scale multi-channel images

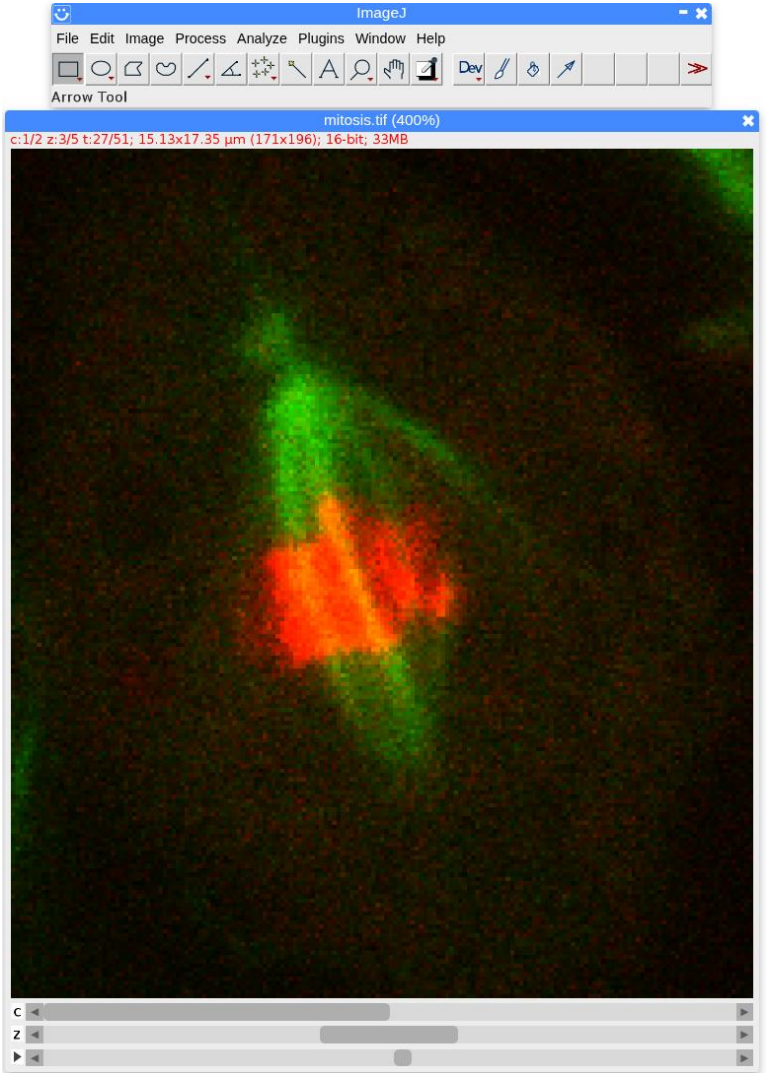
Neuroglancer link



Neuroglancer link



Time-lapse (5D image)



TCZYX

[Imajej.js]

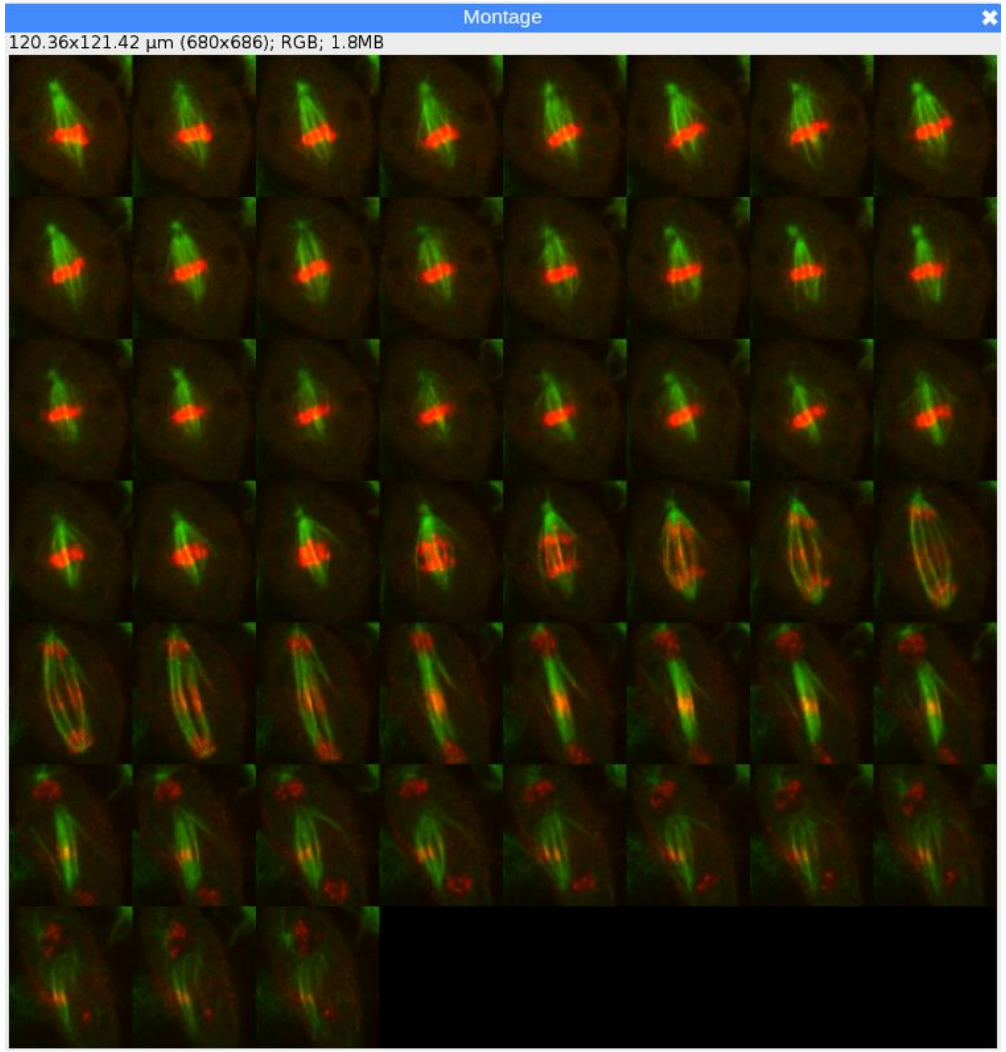
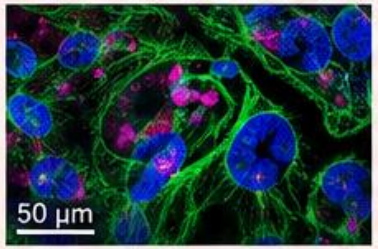


Image compression

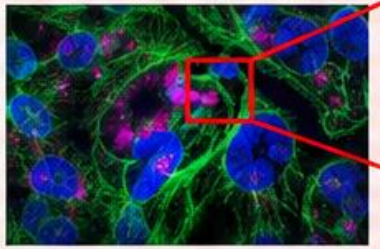
Lossy Compression

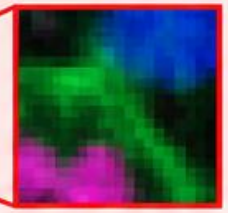
Removes less important information to achieve higher compression.

Original image
(24 MB, uncompressed)



Lossy compressed
(1.2 MB, ~20× smaller)





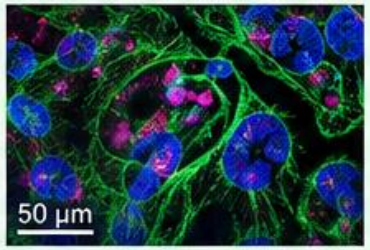
Fine details are lost
(blurred, blocky artifacts)

- **Achieves much smaller file sizes.**
- **Some image quality is permanently lost**
  (e.g., subtle structures, faint signals, sharp edges).
- **Not recommended for quantitative analysis,**
  as pixel intensities can change.
- **Often used for sharing, web display, and publications,**
  where a small amount of loss is acceptable.

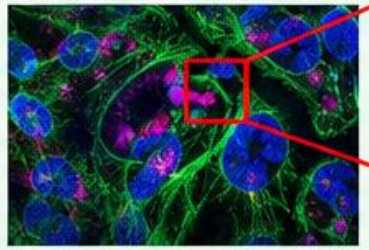
Lossless Compression

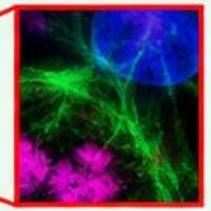
Finds and encodes patterns in the data, without losing any information.

Original image
(24 MB, uncompressed)



Lossless compressed
(8 MB, ~3× smaller)





All details are preserved
(identical to original)

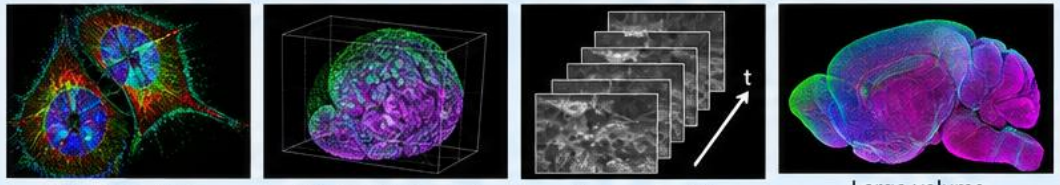
- **Keeps all of the original data** (exact same pixel values).
- **Typically achieves moderate file size reduction**
  (smaller than the original, but usually not as small as lossy).
- **Maintains quantitative accuracy**
  (important for measurements, segmentation, and analysis).
- **Recommended for image analysis, archiving, and data sharing**
  when preserving the original data is important.

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File formats

Microscopy (light microscopy)

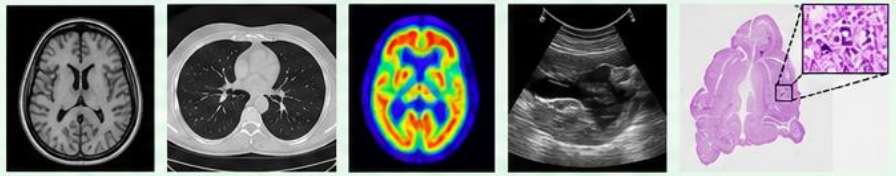
Fluorescence, brightfield, confocal, light-sheet, super-resolution, etc.



2D / time-lapse | 3D / multi-channel | Time series (T) | Large volume (whole organ)

Other biomedical imaging

MRI, CT, PET, ultrasound, histology (WSI), etc.



MRI | CT | PET | Ultrasound | Histology (WSI)

Open formats (widely used)

- **TIFF** — .tif /.tiff — Most common (2D, 3D, time-lapse)
- **OME-TIFF** — .ome.tif — With metadata (OME)
- **OME-Zarr** — .ome.zarr — Multiscale Cloud/HPC (large data)
- **Zarr** — .zarr — Chunked, scalable
- **HDF5** — .h5 /.hdf5 — Large, flexible
- **NIfTI** — .nii /.nii.gz — Common in bioimaging
- **NGFF (Zarr v3)** — .zarr — Next-generation multiscale
- **BTF** — .btf — High-throughput microscopy

Proprietary formats (vendor-specific)

- **CZI** (Zeiss) — .czi
- **LIF** (Leica) — .lif
- **ND2** (Nikon) — .nd2
- **LSM** (Zeiss) — .lsm
- **IMS** (Imaris) — .ims
- **VSI** (Olympus) — .vsi

...

Open formats (widely used)

- **DICOM** — .dcm — Standard for clinical imaging
- **NIfTI** — .nii /.nii.gz — Common for MRI and neuro imaging
- **NRRD** — .nrrd — Flexible, metadata-rich
- **MHD/MHA** — .mhd/.mha — Simple volume format

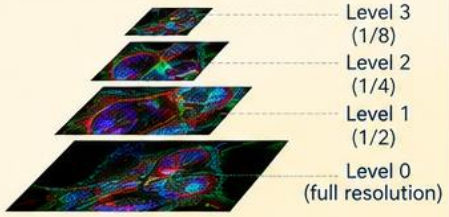
Proprietary formats (vendor-specific)

- **Analyze** — .img /.hdr — Legacy MRI format
- **ECAT** (PET) — .v
- **Siemens** — .ima — MRI/CT (siemens)

...

Multiscale formats

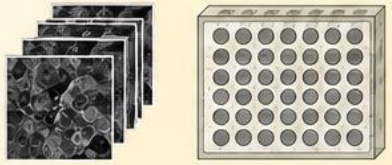
Store multiple downsampled resolutions (image pyramid)



Examples: OME-Zarr, NGFF (Zarr v3), OME-TIFF (pyramid), TIFF (pyramid), etc.

High-throughput formats

Designed for very large datasets (many images, wells, positions, time points)



...

Examples: OME-Zarr, NGFF (Zarr v3), HDF5, BTF, etc.

Bio-Formats

Read, write and convert between 100+ formats

BIO-FORMATS

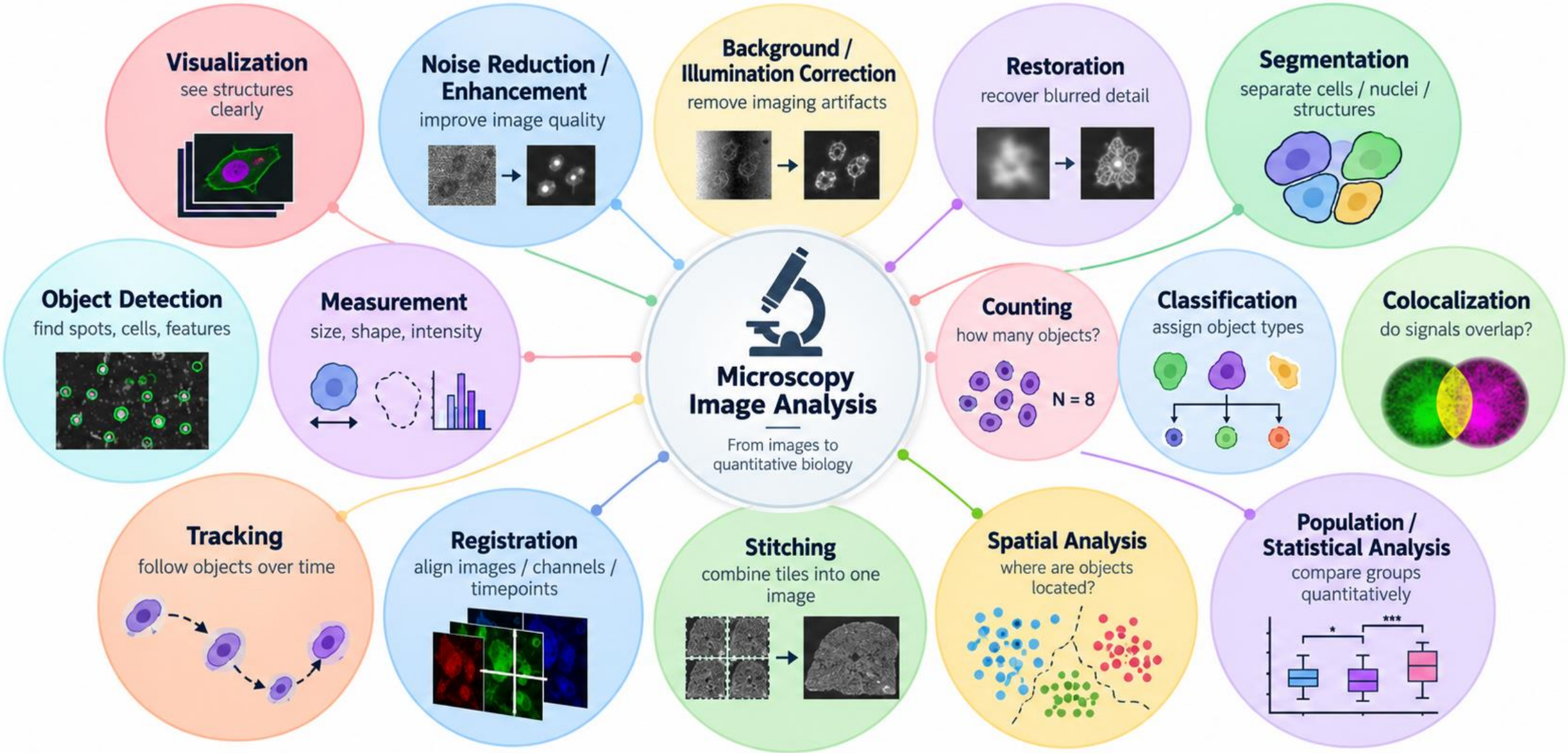
- Supports many microscopy and other biomedical formats
- Converts between formats
- Reads proprietary formats without vendor software
- Access from many tools (Fiji/ImageJ, Python, MATLAB, etc.)

Use in common software tools

- **Fiji / ImageJ** (via Bio-Formats)
- **Python** (e.g. imageio, PyBioFormats)
- **MATLAB** (via Bio-Formats)
- **napari** (via Bio-Formats)
- **QuPath** (via Bio-Formats)
- **Other tools** (e.g. Imaris, CellProfiler, custom workflows)

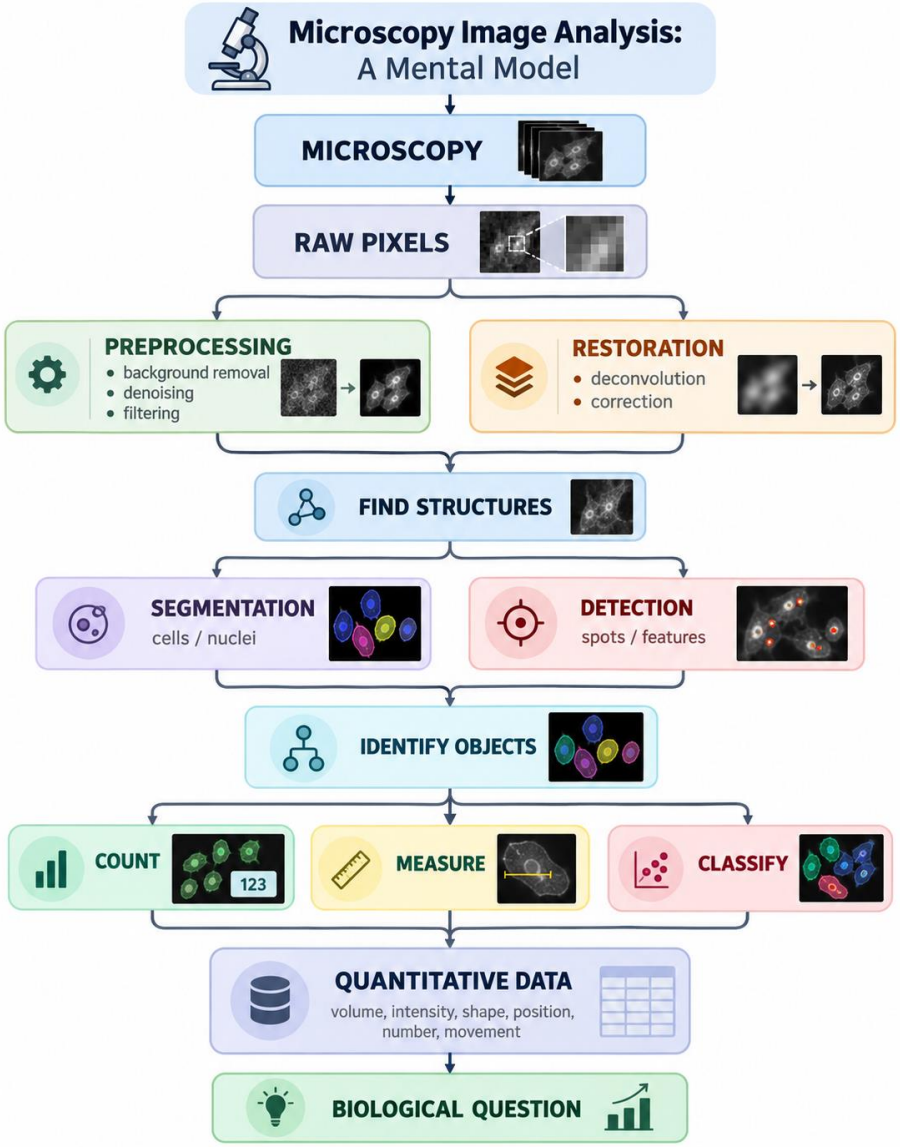
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Recap: Mind Map



Generated by ChatGPT 5

Recap: Full workflow



Generated by ChatGPT 5

Microscopy Image Analysis Software Overview

Major tools in the ecosystem and what they are used for

General-purpose open source

Interactive image processing and analysis



- Fiji / ImageJ
- napari
- CellProfiler
- ilastik
- QuPath

Python / coding tools

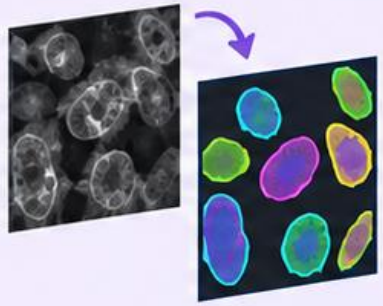
Custom, reproducible analysis workflows



- scikit-image
- OpenCV
- SimpleITK
- NumPy / SciPy
- PyTorch / TensorFlow

AI segmentation & restoration

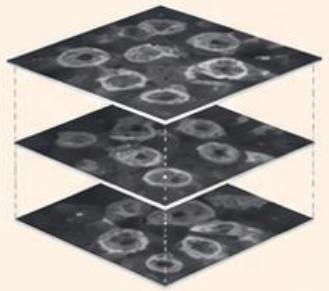
Pretrained models and deep learning



- Cellpose
- StarDist
- Omnipose
- DeepImageJ

Stitching, registration & 3D

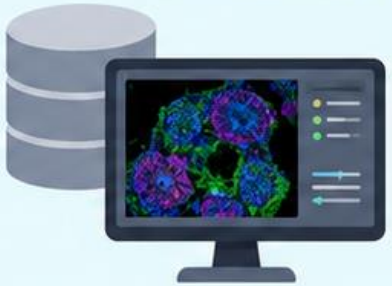
Large volumes, alignment, reconstruction



- BigStitcher
- Elastix
- ANTs
- TrakEM2

Data management & visualization

Formats, storage, viewing, collaboration



- OMERO
- Bio-Formats
- Imaris
- Arivis Vision4D

Vendor / commercial ecosystems

Acquisition plus analysis in microscope workflows



- Zeiss ZEN
- Nikon NIS-Elements
- Leica LAS X
- Evident cellSens

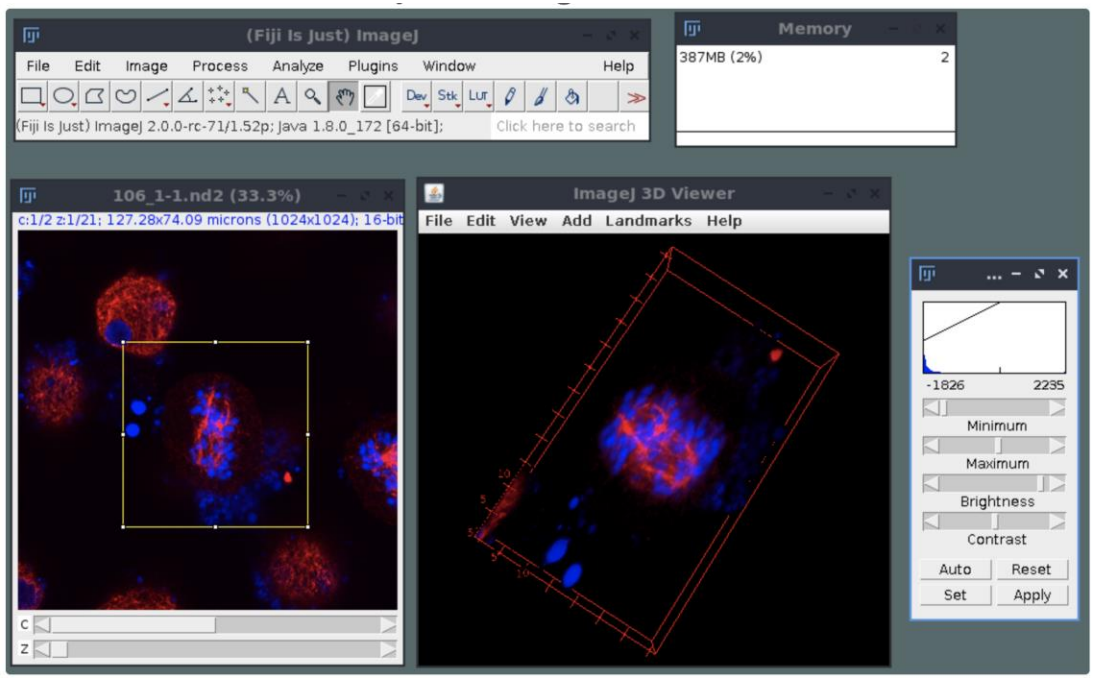
How people often choose tools

- **Need easy GUI →** Fiji, CellProfiler, QuPath
- **Need machine learning →** ilastik, Cellpose, StarDist
- **Need coding flexibility →** Python + napari
- **Need very large 3D / whole-slide data →** BigStitcher, QuPath, Imaris, Arivis

?

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ImageJ (Fiji)



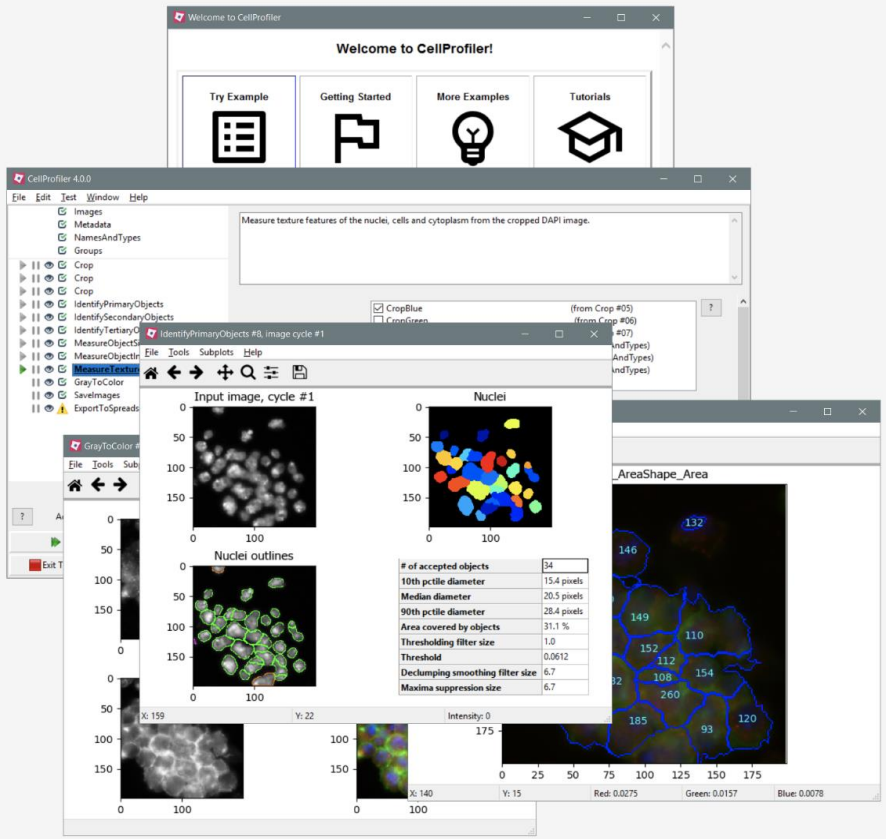
What to know about Fiji:

- classical and versatile
- convenient for all simple operations
- extendable by plugins
- Macro language

Source: https://imagej.github.io/

CellProfiler

CellProfiler™
cell image analysis software

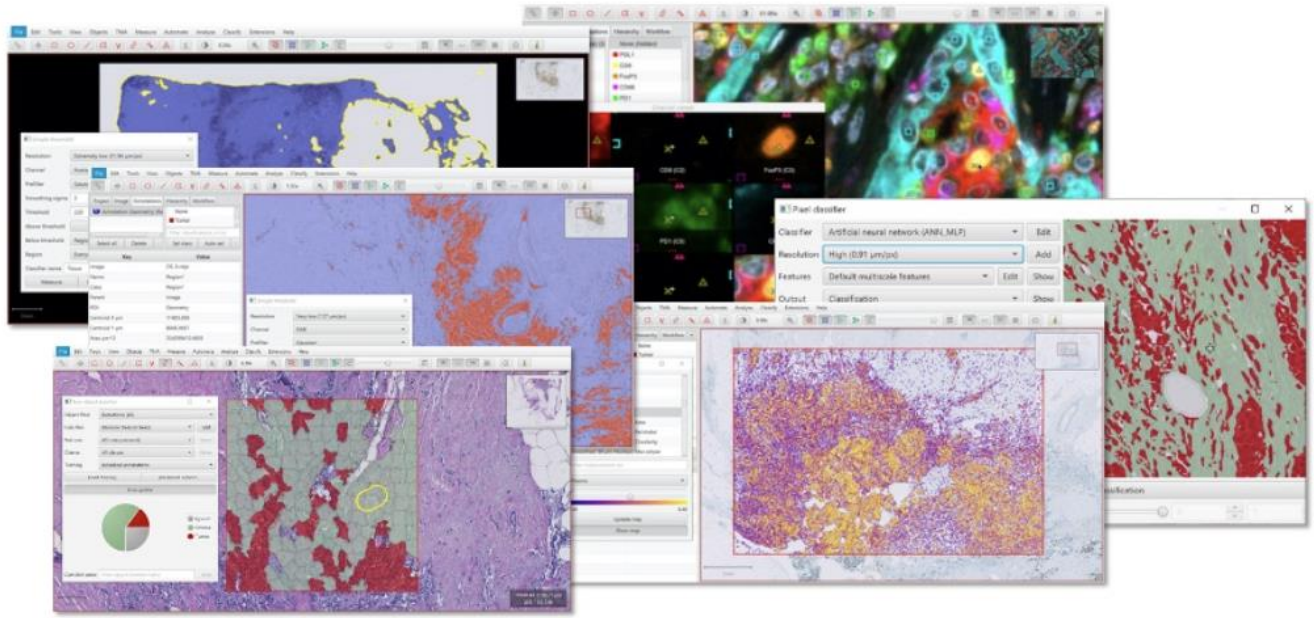


What to know about CellProfiler:

- great for the analysis of 2D fluorescence images of cell cultures – image cytometry
- intuitive with designing medium-throughput analysis (graphical interface to set processing of multiple files)
- sharable analysis pipelines

https://cellprofiler.org/

QuPath



What to know about QuPath:

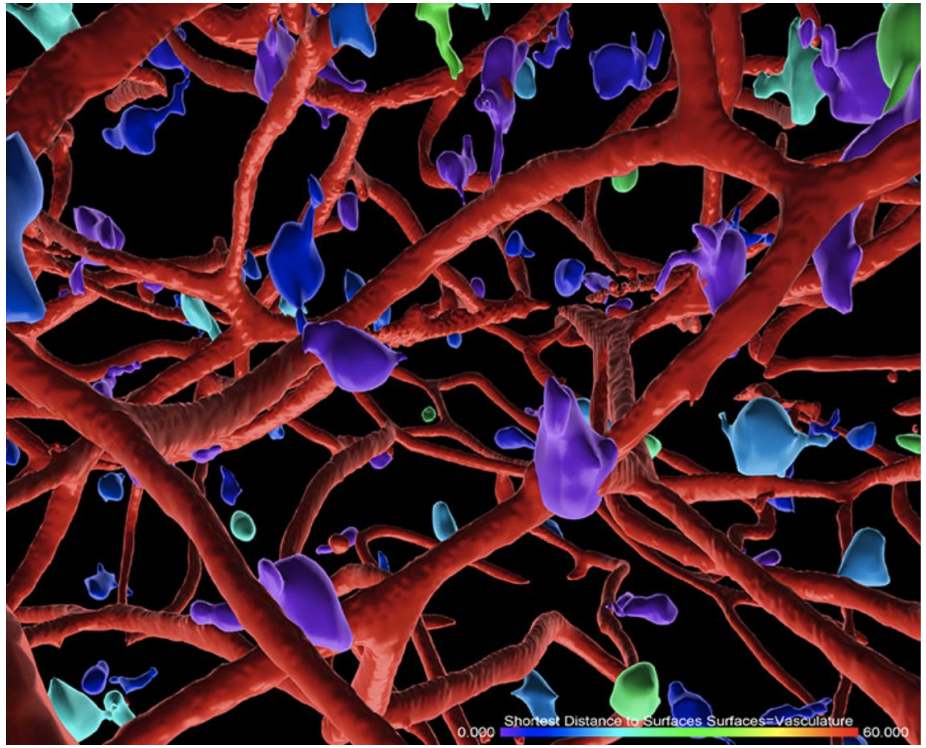
designed for digital pathology (processes well gigantic pyramid images of slides)

contains powerful classifiers of cells and regions

Interactive ML and DL algorithms

https://qupath.github.io/

Imaris

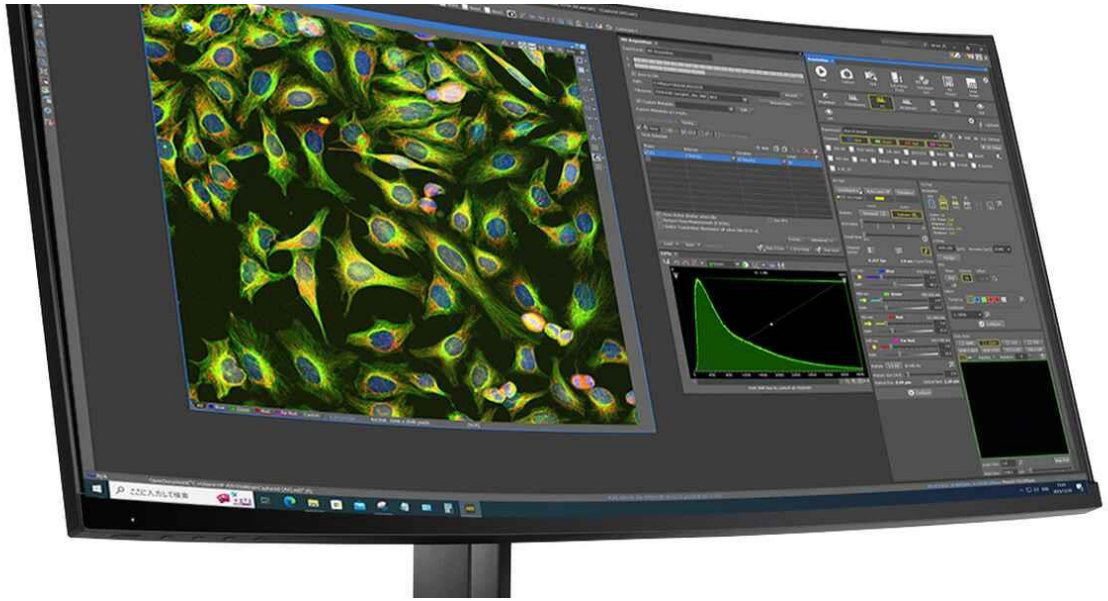


What to know about Imaris:

- proprietary software (access through CBI)
- great surface and volume rendering tools (visualization) for big datasets
- great support for 3D quantitative analysis
- ImarisViewer (only image visualization) available freely

https://imaris.oxinst.com/

NIS-Elements (Nikon)



What to know about Elements:

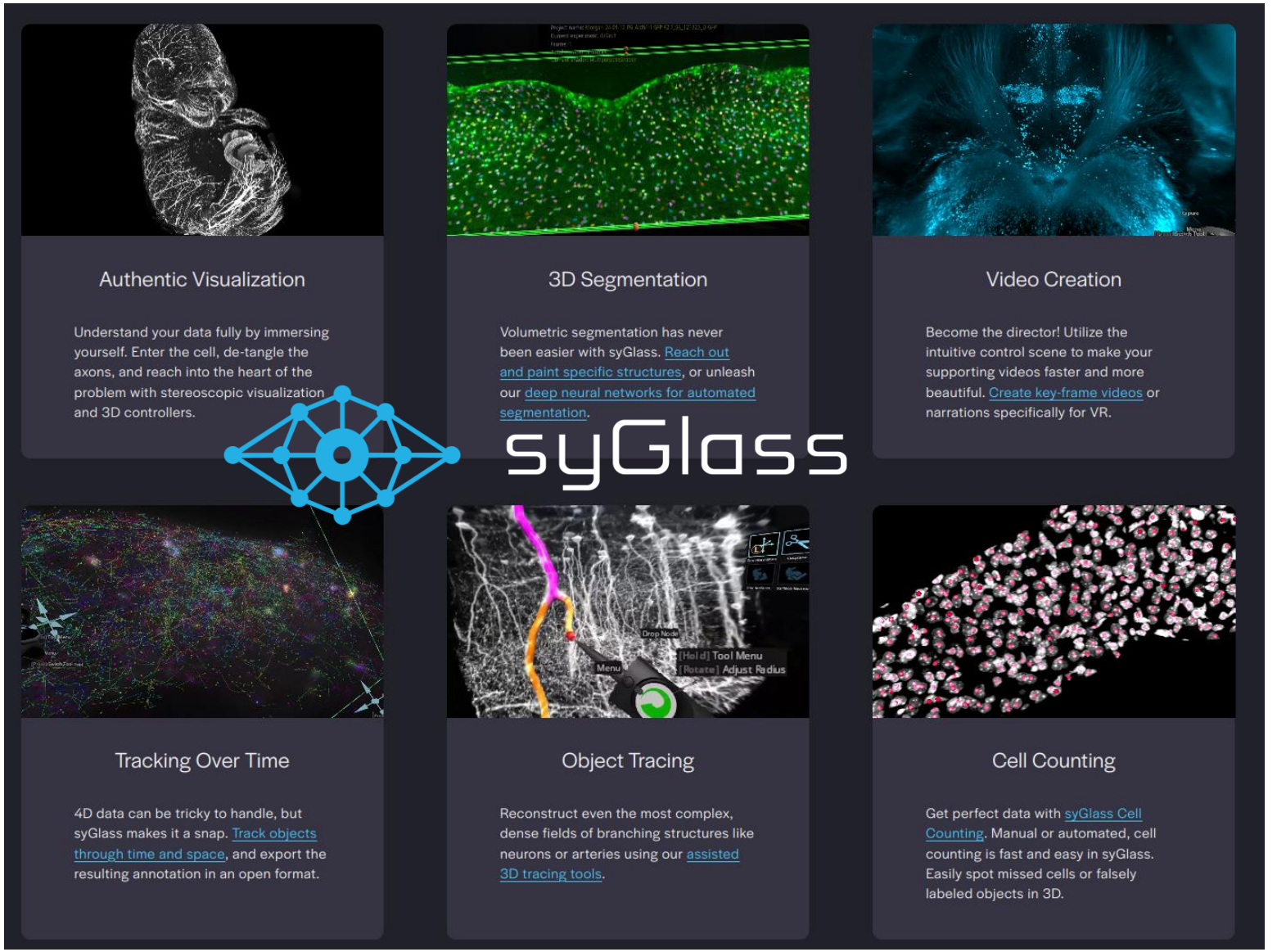
- proprietary software (access through CBI)
- great set of standard and ML image analysis tools packaged in a user-friendly interface
- both 2D and 3D analysis

https://www.microscope.healthcare.nikon.com/products/software/nis-elements

Syglass (virtual reality)

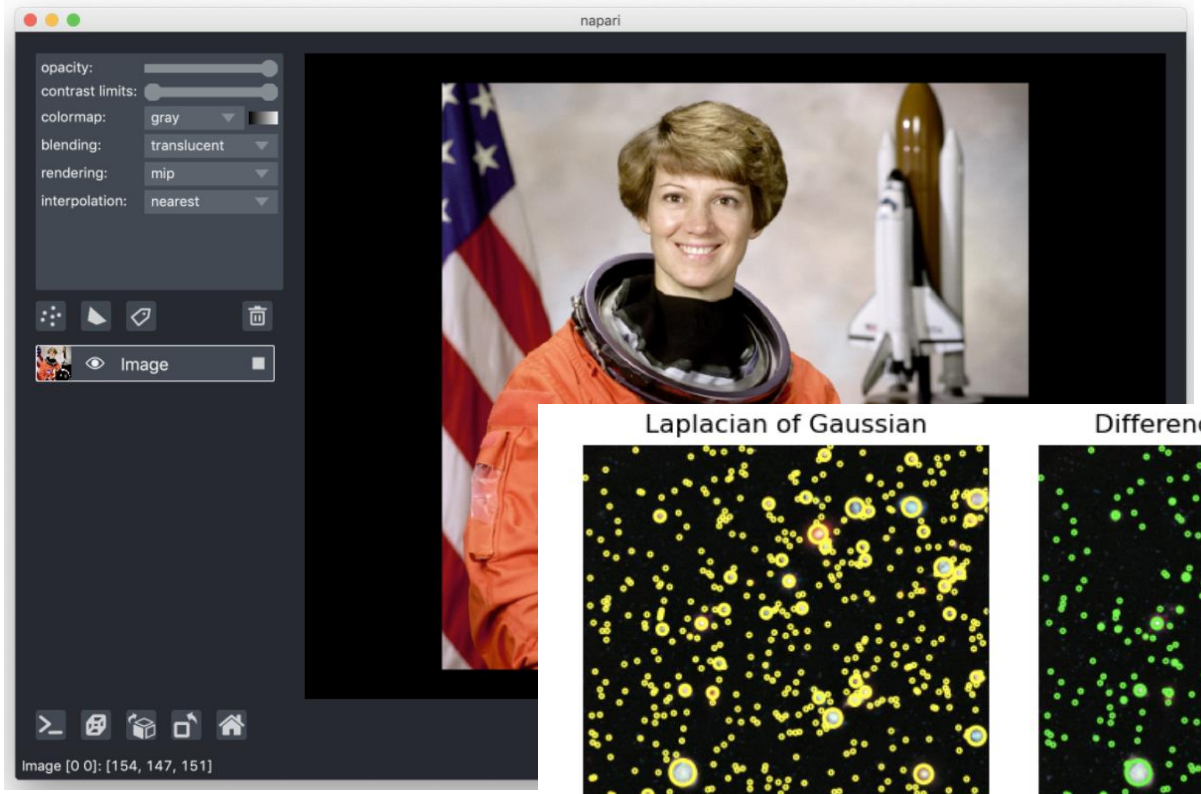






https://www.syglass.io/

Python ecosystem

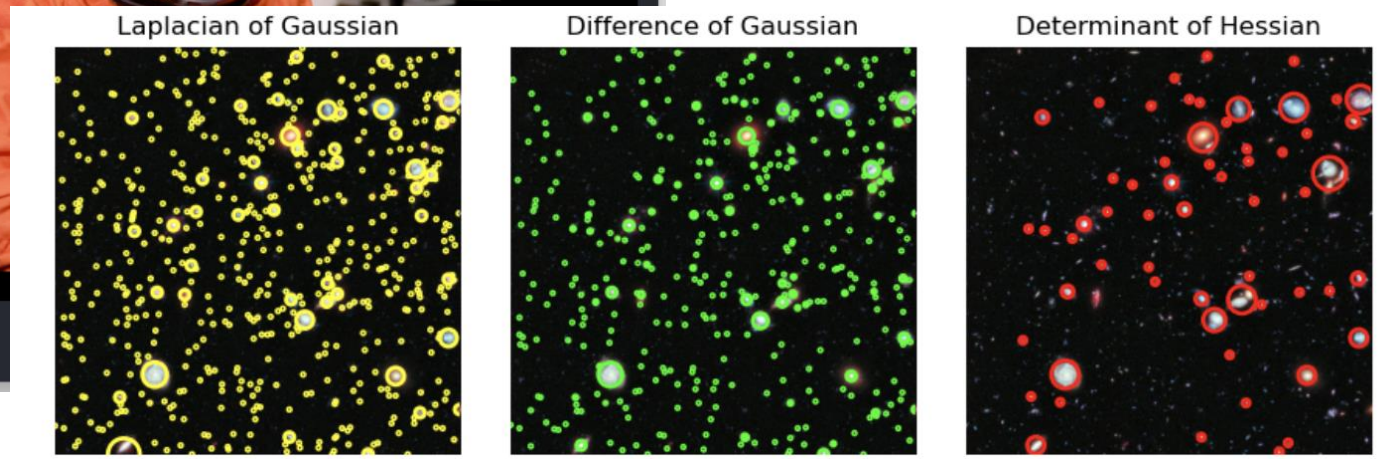


Python

High-level programming language

scikit-image

image processing in python



napari

https://www.python.org/

https://scikit-image.org/

https://napari.org/

Python ecosystem

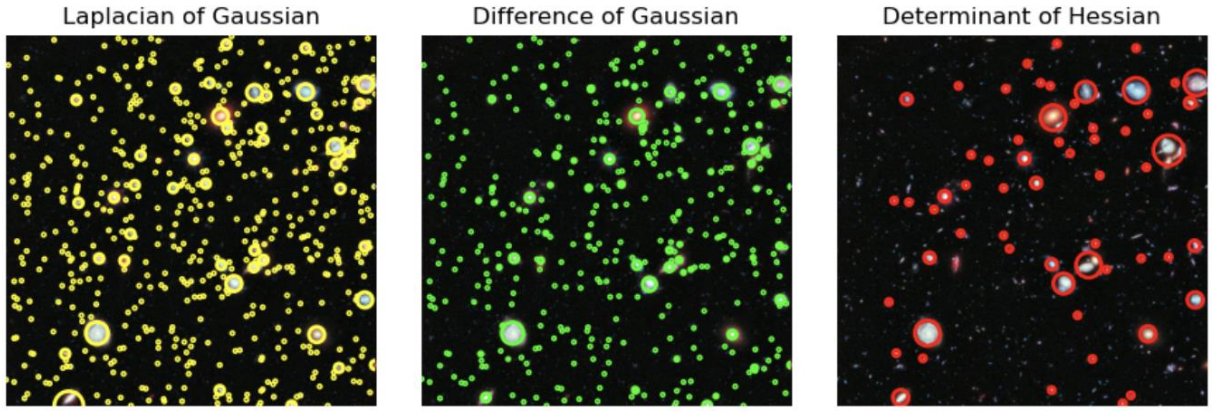
Python

High-level programming language

napari

scikit-image

image processing in python



What to know about image analysis in Python:

- the most popular programming language
- ultimate flexibility
- majority of new algorithms is published in python
- incomparable support for AI

https://www.python.org/
https://scikit-image.org/
https://napari.org/

Image.sc forums

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Related Communities Websites There are several other community forums whose topics overlap with this one: Microforum's focus is hardware, acquisition, and specimen-related aspects of scientific imaging, particularly (but not limited to) theory a… read more		2	15.4k	Feb '20

Today's project

Use Jupyter notebook to work with:

1) Synthetic images
2) Real 2D image (brain slice)
3) Advanced (optional): small 3D image

Real data (from the encephalitis study):
Mouse brain infected by VEEV Aerosol, 96h post infection ->

Research question: what % of the red spots overlap with the green (GFP) signal?





Neuroglancer link